

Bias-Variance-Tradeoff

• Goal: We want to learn $f(x)$ from data.

But, real world data = noisy data.

• $h(x) = \mathbb{E}[t|x] \longrightarrow$ True function

" Average value of t if we repeat the experiment ∞ times?

eg- $h(x) = \sin(2\pi x)$

• If model is very flexible (many basis functions) $\longrightarrow$ noisy fluctuations $\longrightarrow$ overfitting

• If model is very simple (few basis functions) $\longrightarrow$ can't capture true curve $\longrightarrow$ underfitting.

$$\mathbb{E}_{\mathcal{D}}[(f(x; \mathcal{D}) - h(x))^2]$$

This is the average of all possible datasets of size N

Calculation:

$$(f(x; \mathcal{D}) - h(x))^2$$

Add & Subtract the average model (prediction) over datasets:

$$\bar{f}(x) = \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})]$$

$$\Rightarrow (f(x; \mathcal{D}) - h(x)) = (f(x; \mathcal{D}) - \bar{f}(x)) + (\bar{f}(x) - h(x))$$

Squaring both sides:

$$[(f - \bar{f}) + (\bar{f} - h)]^2 = (f - \bar{f})^2 + (\bar{f} - h)^2 + 2(f - \bar{f})(\bar{f} - h)$$

Now take expectation over datasets $\mathcal{D}$.

$$\mathbb{E}[(f - h)^2] = \mathbb{E}[(f - \bar{f})^2] + \underbrace{\mathbb{E}[(\bar{f} - h)^2]}_{\text{doesn't depend on dataset } \mathcal{D}} + 2\mathbb{E}[(f - \bar{f})(\bar{f} - h)]$$

$$\bar{f}(x) = \mathbb{E}_{\mathcal{D}}[f(x; \mathcal{D})]$$

$h(x) \rightarrow$ true function

$\therefore$ They're both constants w.r.t $\mathcal{D}$.

$$\Rightarrow \mathbb{E}[(f - \bar{f})^2] + (\bar{f} - h)^2 + 2(\bar{f} - h)\mathbb{E}[(f - \bar{f})] \rightarrow 0$$



So, we get:

$$\mathbb{E}[(b-h)^2] = (\bar{b}-h)^2 + \mathbb{E}[(b-\bar{b})^2]$$

$\Rightarrow$

$$= \underbrace{(\mathbb{E}[b]-h)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(b-\mathbb{E}[b])^2]}_{\text{Variance}}$$

